Anomaly Segmentation for Autonomous Driving

Mask-Based Architectures with a Lightweight MLP Anomaly Head

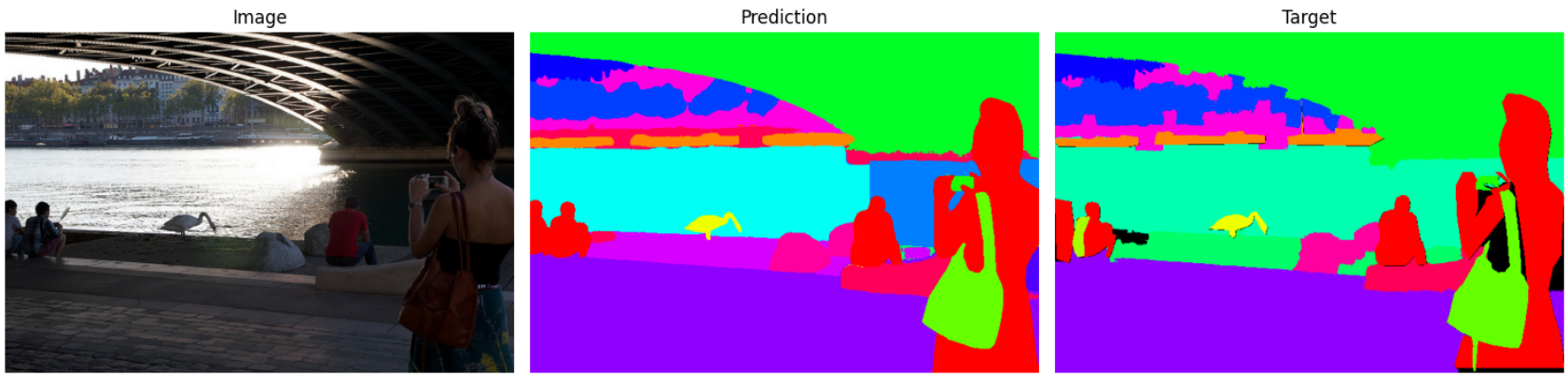
Buttu Danilo 353244
Squeglia Simona 358678
Vita Raffaele 362747
Viltono Federica 364417

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Introduction

- **Motivation**: Safety in autonomous driving requires models to recognize unknown objects that were never seen during the training phase.
- **Semantic Segmentation**: This is the process of assigning a category label to every individual pixel in an image.
- **Goal**: Enhance robustness against novel anomalies. Evaluated across diverse road scene benchmarks: SMIYC, Fishyscapes, and Road Anomaly.
- **Our Approach**: Benchmark pixel-based and mask-based systems using post-hoc scoring methods. Then introduce a parallel binary MLP head trained on synthetic outlier exposure (COCO objects pasted on Cityscapes backgrounds) to learn a direct anomaly score.



Model Architectures: ERFNet

- **Pixel-Wise Baseline:** A fully convolutional encoder-decoder network designed for per-pixel semantic classification.
- **Efficiency:** Uses factorized convolutions to reduce parameters while preserving the receptive field.
- **Output Space:** Maps resized inputs into pixel-level logits across 20 categories: 19 Cityscapes classes + 1 void class.
- **Anomaly Scoring:** Trained under a closed-world assumption. It is forced to map unknown anomalies into known classes, relying on post-hoc uncertainty metrics for anomaly detection.



Model Architectures: EoMT

- **Mask-Based Paradigm:** Abandons per-pixel classification. Employs learnable object queries to simultaneously predict binary masks and class probability vectors.
- **Foundation Model Backbone:** Leverages a frozen DINOv2 vision transformer to extract rich, open-world semantic features without retraining.
- **Resolution Strategy:** Processes overlapping crops at native resolution to prevent geometric distortions and preserve small obstacles.
- **Logit Aggregation Formula:** Computes final pixel-level semantic logits by blending mask probabilities with class predictions:

$$\text{logits}_c(x, y) = \sum_q \sigma(\text{mask}_q(x, y)) \cdot \text{softmax}(\text{class}_q)_c$$



Baseline Evaluation: ERFNet

1. **Preprocessing:** Initializes the model with pretrained weights using 512×1024 resized input images, keeping pixel values in the range $[0, 255]$.
2. **Forward pass:** The resized image is passed to the model and it returns a tensor of logits of shape $[1, 20, 512, 1024]$.
3. **Anomaly score computation:** Selects a post-hoc method as input:
   - **MSP:** $1 - \max_c \text{softmax}(\text{logits})_c$
   - **Max Logit:** $-\max_c \text{logits}_c$
   - **Max Entropy:** $-\sum_c p_c \log(p_c + \varepsilon)$, with $p = \text{softmax}(\text{logits})$
4. **Ground truth alignment:** The ground truth mask is resized and remapped into a binary anomaly mask (0, 1 or 255).
5. **Metric accumulation:** Metrics are computed on concatenated valid pixels:
   - **AuPRC**: Higher values mean better anomaly detection
   - **FPR@95TPR**: Lower values mean fewer false alarms with a 95% treshold

Baseline Evaluation: EoMT

1. **Windowing:** Splits the original image into overlapping crops of a fixed size.
2. **Forward Pass:** Converts crops to float, normalizes pixels and feeds them to the network to obtain mask logits and class logits.
3. **Compute per-crop semantic logits:** Combines mask and class logits via a sigmoid-softmax multiplication into a $[B, C, H_{\text{crop}}, W_{\text{crop}}]$ tensor.
4. **Revert Window Logits**: Logits are stitched back to the initial full-resolution image, averaging overlapping regions to get a single tensor of shape $[1, C, H_{\text{orig}}, W_{\text{orig}}]$.
5. **Anomaly Score Computation:** Selects a post-hoc baseline method as input:
   - **MSP / Max Logit / Max Entropy:** Computed as ERFNet.
   - **RbA:** $-\sum_{c=1}^{C} \tanh(\text{logits}_c)$
6. **Ground Truth Alignment and Metric Accumulation:** Resizes the ground truth mask to create a binary anomaly mask, evaluating performance over accumulated valid pixels via the AuPRC and FPR@95TPR metrics.

Baseline Results

MODEL EVALUATION											
		SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
model	method										
ERFNET	**MSP**	29,10	62,55	2,71	65,22	1,75	50,59	7,47	41,84	12,42	82,58
	Max Logit	38,32	59,34	4,63	48,44	3,30	45,49	9,50	40,30	15,58	73,25
	Max Entropy	30,97	62,66	3,04	65,91	2,58	50,16	8,84	41,55	12,67	82,75
EoMT	**MSP**	66,03	34,28	89,70	0,34	17,60	19,70	62,53	28,58	66,97	19,73
	Max Logit	65,92	34,64	89,68	0,34	17,60	19,49	62,51	29,71	66,54	19,73
	Max Entropy	69,08	34,08	89,84	0,32	19,63	19,58	60,47	28,88	70,25	19,21
	RbA	63,80	84,64	88,42	0,67	17,86	12,98	62,83	22,77	66,10	17,08

- The results were tested on 5 datasets: RoadAnomaly21, RoadObstacle21, FS Lost&Found, FS Static and RoadAnomaly.
- Max Logit achieves the best performance.
- EoMT is definitely outperforming ERFNet across all scenarios and metrics.
- EoMT leverages the RbA method to achieve superior uncertainty estimation.

Temperature Scaling and mIoU Invariance

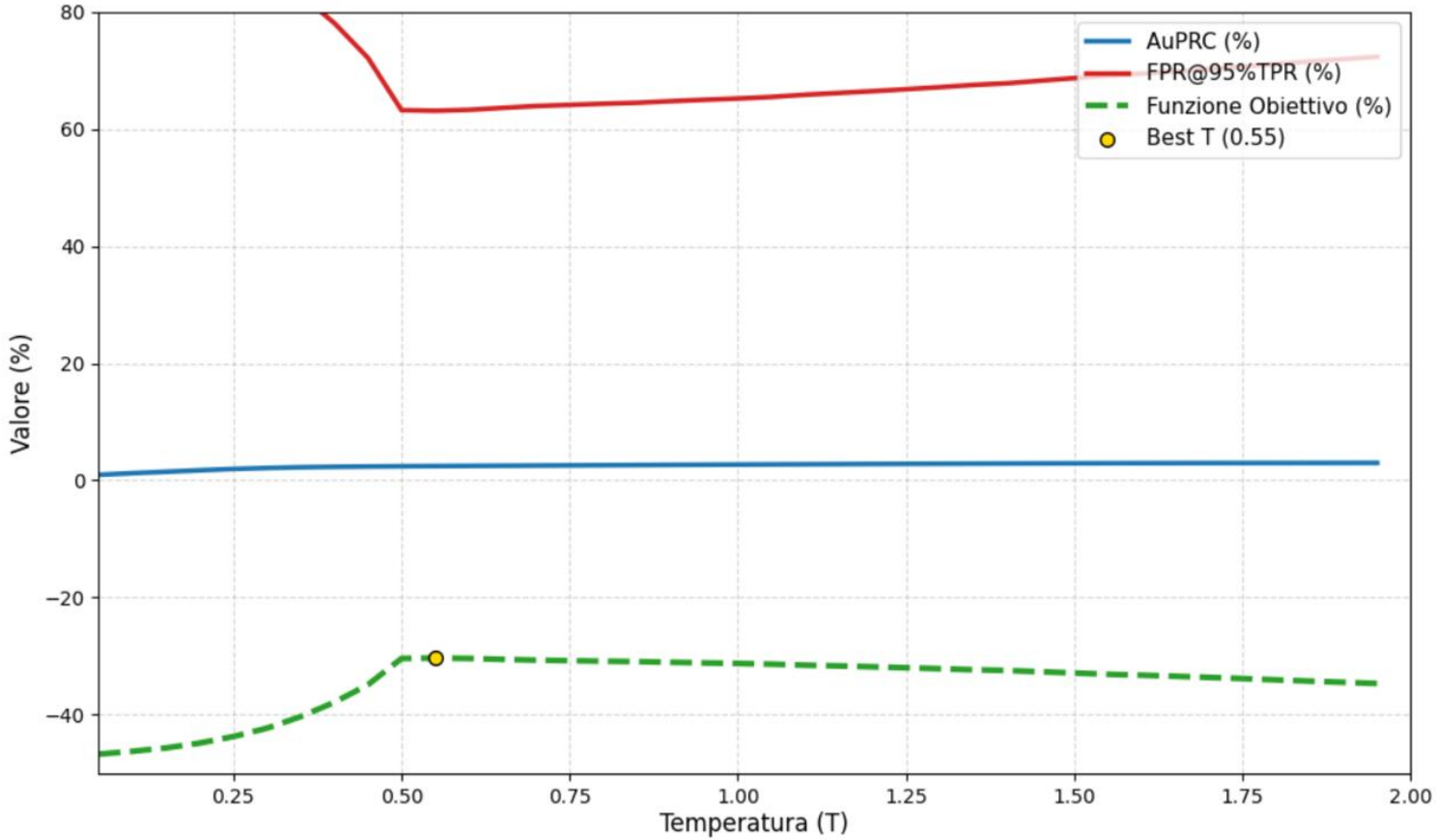
- **Temperature Scaling**: Logits are scaled by a positive scalar to refine the model's probability distribution without altering the final class predictions. It is applied to improve the anomaly detection results.
- **mIoU**: The metric measures how accurately the model segments and recognizes the known, standard classes (like roads, cars or pedestrians). It remains constant as t varies.

TEMPERATURE SCALING											
		SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
method	mIoU	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP (t=0.5)	72,17	27,06	62,73	2,42	63,23	1,28	66,74	6,60	43,48	12,19	82,02
MSP (t=0.75)	72,17	28,16	62,49	2,57	64,13	1,49	51,76	6,99	42,50	12,32	82,31
MSP (t=1)	72,17	29,10	62,55	2,71	65,22	1,75	50,59	7,47	41,84	12,42	82,58
MSP (t=1.1)	72,17	29,40	62,65	2,76	65,87	1,86	50,18	7,69	41,62	12,46	83,73
MSP (best t)	72,17	29,89	63,01	2,48	63,28	3,26	44,87	9,83	41,47	12,19	82,02
		t = 1.30		t = 0.6		t = 5		t = 2,2		t = 0,5	

		SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
method	mIoU	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP (t=0.5)	79,73	65,98	34,28	89,69	0,34	17,59	19,70	62,52	28,58	66,88	19,79
MSP (t=0.75)	79,73	66,00	34,28	89,70	0,34	17,59	19,70	62,53	28,58	66,98	19,75
MSP (t=1)	79,73	66,06	34,28	89,70	0,34	17,60	19,70	62,53	28,58	66,07	19,73
MSP (t=1.1)	79,73	66,08	34,28	89,70	0,34	17,60	19,70	62,54	28,58	67,10	19,73
MSP (best t)	79,73	66,37	34,28	89,71	0,34	17,62	19,7	62,55	28,58	67,5	19,71
		t = 4		t = 2		t = 1.5		t = 2,1		t = 4,3	

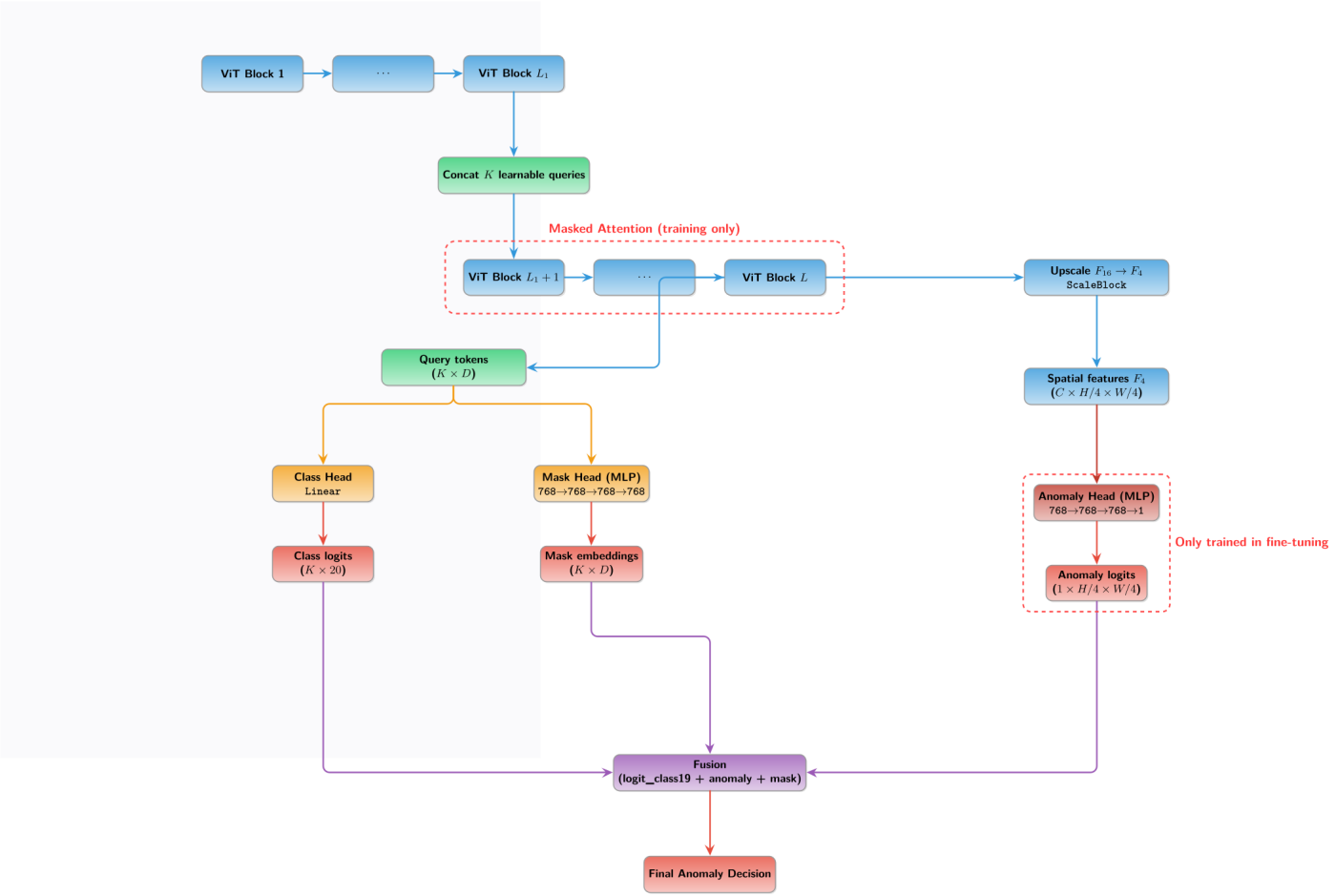
Temperature Scaling

- **Optimal Temperature Selection**: The best t is identified by maximizing the objective function: Objective $= \frac{\text{AuPRC} - \text{FPR95}}{2}$.



Extension: Lightweight MLP Anomaly Head

- A parallel **binary MLP head** attached to frozen EoMT encoder.
- Trained exclusively on ad-hoc synthetic anomaly masks (binary cross-entropy).
- Outputs a direct anomaly score per pixel.
- Inference: fuse MLP score with native EoMT anomaly score.



Synthetic Dataset Creation

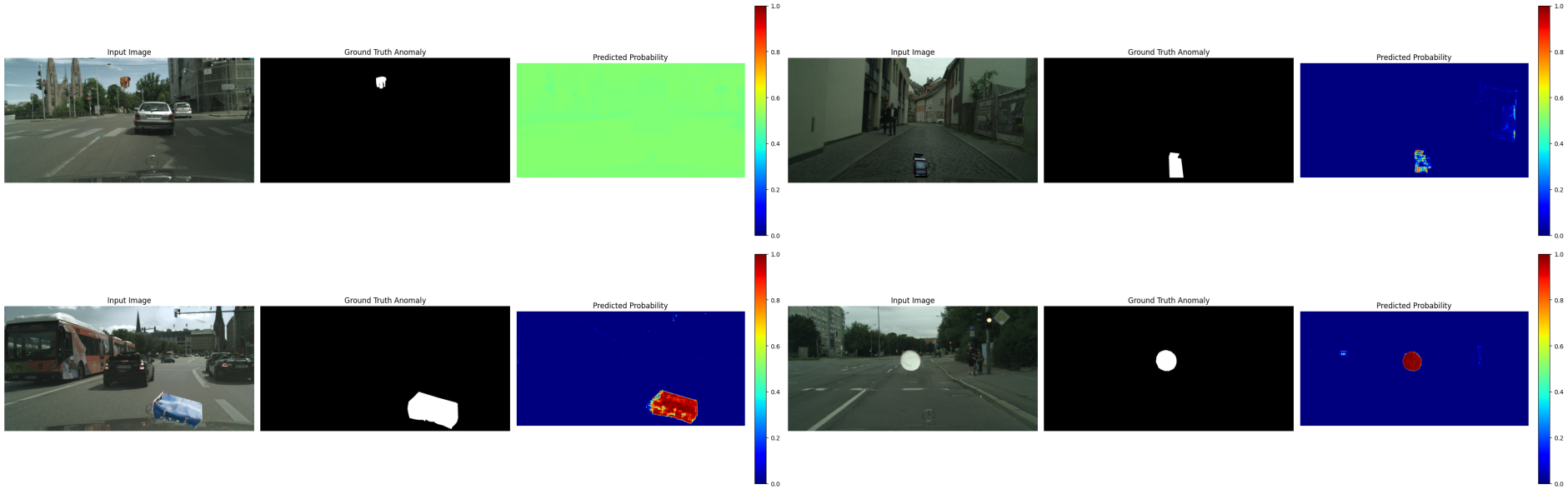
- Copy-paste from COCO 2017 onto Cityscapes backgrounds.
- Exclude Cityscapes-like categories (person, car, traffic light, etc.).
- Use objects like teddy bear, suitcase, sports ball, banana, umbrella as anomalies to detect.



Examples of images created for training.

Training Details

- **MLP head:** 3 layers of dimension 768 with GELU as activation functions.
- **Epochs:** 50, batch size 8 (accumulated).
- **Input resolution:** 1024×1024 (feature map upscaled to 256×256).
- **Loss:** Binary cross-entropy with synthetic anomaly mask.



Examples of model predictions during training on different scenes.

How to fuse EoMT anomaly score & MLP anomaly score?

Logic behind each fusion strategy:

- **Max Fusion:** Takes the most pessimistic (highest) anomaly evidence from either branch.

$$S_{hybrid} = \max(S_{mlp}, S_{classic})$$

- **Mean Fusion:** Averages the evidence from both branches to smooth out false positives.

$$S_{hybrid} = \frac{S_{mlp} + S_{classic}}{2}$$

- **Dot Product Fusion:** Requires high confidence from *both* branches simultaneously to register a strong anomaly.

$$S_{hybrid} = S_{mlp} \cdot S_{classic}$$

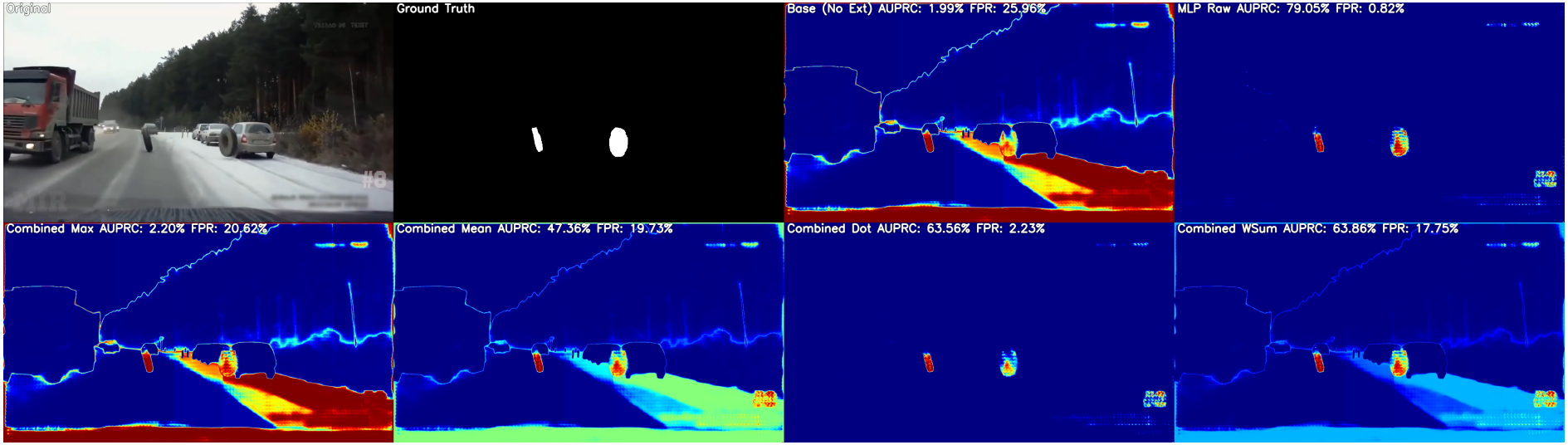
- **Weighted Sum:** Heavily favors the explicitly learned MLP features while retaining a regularization signal from the semantic head.

$$S_{hybrid} = 0.7 \cdot S_{mlp} + 0.3 \cdot S_{classic}$$

- **Raw (MLP-Only):** Bypasses fusion entirely to evaluate the independent detection capabilities of the trained MLP extension.

$$S_{hybrid} = S_{mlp}$$

Visual Comparison of Fusion Strategies



Grid of anomaly maps for each fusion method.

Results – Crop Size 640×640 (Extension vs Baseline)

Combine	Best Method	SMIYC RA-21 AuPRC / FPR95	Road Anomaly AuPRC / FPR95
Baseline EoMT	**MSP**	**66.03 / 34.28**	**66.97 / 19.73**
raw	-	71.90 (↑) / 65.20 (↑)	89.70 (↑) / 4.07 (↓)
max	Max Entropy	71.60 (↑) / 30.28 (↓)	72.20 (↑) / 16.93 (↓)
mean	Max Entropy	75.76 (↑) / 30.32 (↓)	88.50 (↑) / 15.79 (↓)
dot	Max Logit	**77.06** (↑) / 54.42 (↑)	**91.34** (↑) / **3.88** (↓)
weighted_sum	RbA	74.80 (↑) / 30.30 (↓)	88.03 (↑) / 12.33 (↓)

- **Compared to Baseline**: The extension drastically improves performance on Road Anomaly. With *dot* fusion and Max Logit, AuPRC goes from 66.9% to 91.3% and FPR95 drops from 19.7% to 3.8%.
- **Trade-off on RA-21**: While most methods improve RA-21 metrics overall, *raw* and *dot* combinations suffer a significant FPR95 degradation compared to the baseline.

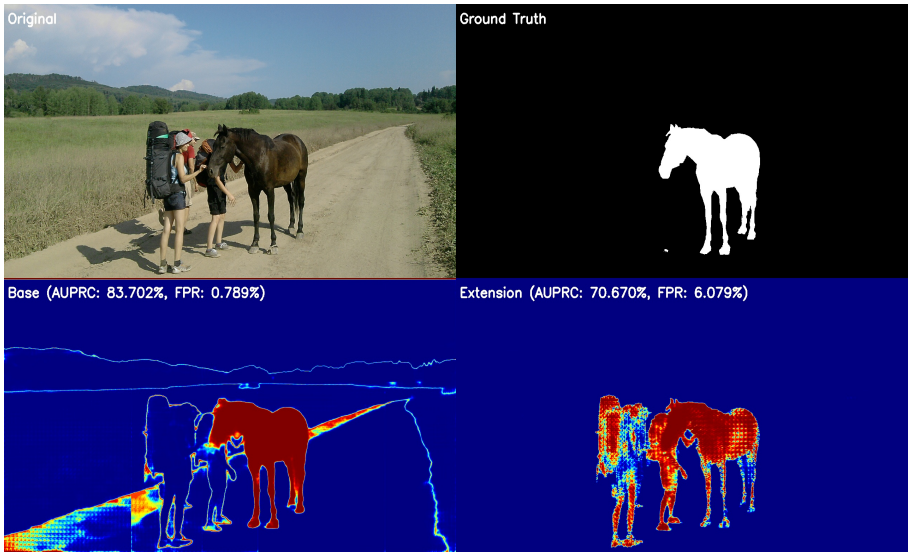
Results – Crop Size 1024×1024 (Comparison vs 640×640)

Combine	Best Method	SMIYC RA-21 AuPRC / FPR95	Road Anomaly AuPRC / FPR95
raw	-	70.37 (↓) / 70.88 (↑)	89.89 (↑) / 4.02 (↓)
max	Max Entropy	69.19 (↓) / 37.89 (↑)	74.49 (↑) / 12.53 (↓)
mean	Max Entropy	75.01 (↓) / 37.95 (↑)	89.95 (↑) / 8.90 (↓)
dot	Max Logit	75.97 (↓) / 62.36 (↑)	**92.13** (↑) / **3.56** (↓)
weighted_sum	RbA	73.00 (↓) / 56.27 (↑)	89.63 (↑) / 8.23 (↓)

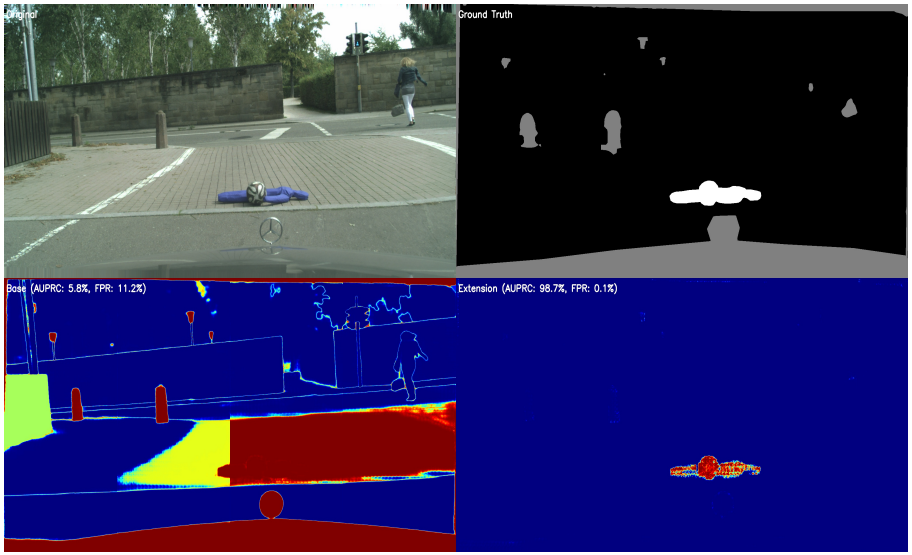
- **Crop Size Impact**: A clear trend emerges at 1024x1024. All methods consistently improve on Road Anomaly (higher AuPRC, lower FPR95), but uniformly worsen on RA-21 compared to 640x640.
- **Best Overall**: The *dot* fusion combined with **Max Logit** at 1024×1024 achieves the best results on Road Anomaly (AuPRC 92.13%, FPR95 3.56%).

Comparison with Baseline EoMT

- Baseline EoMT (Max Entropy) on Road Anomaly: AuPRC 5.8%, FPR95 11.2%.
- Our extension (dot fusion, 1024x1024 crops size): AuPRC 98.7%, FPR95 0.1%.
- **Significant improvement** – learned outlier exposure works.



In this contex the MLP head mistakes the people for anomaly.



In this one is able to detect only the obsticle.

Conclusion & Future Work

Conclusion:

- Mask-based EoMT strongly outperforms pixel-based ERFNet for anomaly segmentation.
- Temperature scaling improves anomaly metrics without harming semantic mIoU.
- Proposed lightweight MLP anomaly head (trained on synthetic copy-paste data) gives **state-of-the-art results**: 92% AuPRC on Road Anomaly.
- Larger sliding-window crops (1024×1024) and advanced fusion strategies (dot product, weighted sum) significantly boost detection of unknown objects (debris, animals, unconventional obstacles).

Future Work / Further Experiments:

- Adaptive crop selection based on image content.
- Semi-supervised refinement with real anomaly examples.
- Extend to panoptic anomaly segmentation.
- **Reconstruction-based architectures:** Train an MLP+CNN to reconstruct only in-distribution scenes; high reconstruction error at test time signals anomalies (unsupervised paradigm that could be ensembled with our explicit outlier exposure).